

- *White agriculture*: system of agriculture based on substantial use of microorganisms, particularly fungi.
- *One-straw Revolution*: system of natural farming without ploughing, chemical fertilizers, weeding and chemical pesticides and herbicides.
- Promotion for animal husbandry.
- Efficient management of water resources.

This could help address the issues of climate change and agriculture and would further increase the productivity in perpetuity without ecological harm. Consequently, new agriculture policy of India aims at sustainable agriculture, which is popularly called 'second green revolution' or 'Evergreen Revolution'.

Check Your Progress 4

- 1) Do you agree with the view that green revolution lost its steam and India needs another green revolution or evergreen revolution? Give adequate justification in support of your answer.

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13.6 TRANSFORMATION OF INDIAN AGRICULTURE THROUGH AN UMBRELLA PROGRAMME OF DOUBLING OF FARMERS' INCOME (DFI) FROM 2015 TO 2022

Past strategy for development of the agriculture sector in India focused primarily on raising agricultural output and improving food security. This strategy involved: (a) an increase in productivity through better technology and varieties, and increased use of quality seed, fertilizer, irrigation and agro-chemicals, (b) incentive structure in the form of remunerative prices for some crops and subsidies on farm product, (c) public investments in and for agriculture, and developing facilitating institutions. This strategy paid dividends as the country was able to address severe food shortage that emerged during 1960s:

- During the period 1965-2015, since the adoption of green revolution, India's food production multiplied 3.7 times while the population multiplied by 2.55 times.
- The net result has been a 45 per cent increase in per person food production, which has made India not only self-sufficient in foodgrains at aggregate level but also a net food exporting country.

This strategy did not recognise the need to raise farmers' income and thus no direct action for farmers' welfare. NSSO data on Household Consumption Survey, 2011-12 reveal that more than 1/5th of rural households with self-employment in agriculture have income below the rural poverty line. During 1980s, farm income

per cultivator was just 34 per cent of the income of non-agricultural worker, which further worsened to 25 per cent in 1993-94. Low level of absolute income of farmers and rising disparity between the farmers' income and non-farm workers resulted in widespread emergence of farmers' distress during 1990s. Low and highly fluctuating farm income has been detrimental to farmers and farm investment, and thus forcing young people leaving agriculture. Realising the need to pay special attention to the plight of farmers, the goal to double the farmer's income by 2022-23 has been set by the government.

13.6.1 Prime Minister's Multi-dimensional Seven Points Strategy

Prime Minister's multi-dimensional seven-point strategy includes:

- 1) Emphasis on irrigation along with end-to-end solution on creation of resources for 'More crop per drop'.
- 2) Provision of quality seeds and nutrients according to the soil quality of each farm.
- 3) Large investments in warehouses and cold chains to prevent post-harvest losses.
- 4) Promotion of value addition through food processing.
- 5) Implementation of National Agricultural Markets and e-platforms (e-NAM) to eliminate shortcomings of all the 585 centres.
- 6) To mitigate the risk, introduction of crop insurance scheme at a lower cost to the farmers.
- 7) Promotion of allied activities such as dairy, animal husbandry, poultry, bee-keeping, horticulture, and fisheries.

13.6.2 Policy Initiatives for Enhancing the Farmers' Output per Hectare: Improving Cropping Intensity

In consonance with crucial sources for raising the cropping intensity, two major flagship schemes launched immediately after announcement of DFI include (a) Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) and (b) Pradhan Mantri Fasal Bima Yojana (PMFBY)

Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)

With almost 54 per cent of the net sown area in the country rain-fed, availability of irrigation facility will ensure access to some means of protective irrigation for all agricultural farms in the country. To this effect Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) has been approved (in July 2015) and formulated with the vision of extending the coverage of irrigation 'Har Khet Ko Pani' and improving water use efficiency 'More crop per drop'.

Pradhan Mantri Fasal Bima Yojana (PMFBY)

Small and marginal farmers holding (86.3 per cent) with 47.3 per cent cropped area dominates India's agriculture. India's primary failure has been its inability to capitalise on technology and efficient agricultural practices, which can ensure surpluses despite small landholdings. Another reason for India's farmers needing insurance is that commercialisation of agriculture requires an increase in credit

needs, but most small and marginal farmers cannot avail credit from formal institutions due to the massive defaults caused by repeated crop failure. Besides, due to climate change higher incidence of extreme weather events aggravates agrarian distress and if this risk is mitigated, farmers will be motivated to make investment in agriculture.

In April, 2016, the government of India had launched PMFBY after rolling back the earlier insurance schemes viz. National Agriculture Insurance Scheme (NAIS), Weather-based Crop Insurance scheme and Modified National Agricultural Insurance Scheme (MNAIS). The PMFBY was created to target 50 per cent of all farmers, with the promise of compensation in case of crop loss.

13.6.3 Policy Initiatives for Crop Diversification and Enhancing Value Realisation

To enhance the farmers' income one of the alternatives is that farmers may be motivated to diversify the cropping pattern from cultivation of cereals such as wheat and rice to high value crops such as fruits and vegetables, commercial crops, spices etc. The idea behind Operation Green is to double the income of farmers by end of 2022 through changing the cropping pattern.

Basic Principles of Operation Green Strategy

- *Principle 1:* To establish the link between farmers and major organised retailers to begin with and thereafter allow them to access the retail markets. The real challenge is to find the right markets that can give them remunerative prices on a sustainable basis. Farmers can be organised in Farmers Producers Organisations (FPOs). NABARD and Small Farmers' Agri-Business Consortium (SFAC) together have about 3,000 FPOs, which could be the starting points for the aggregation of commodities, assaying, sorting, grading, and even packing with bar codes, reflecting their traceability. It is targeted to form at least 10,000 FPOs by 2022.
- *Principle 2:* The existing Agricultural Produce Market Committee (APMC) system of marketing does not allow FPOs to sell their produce directly to the private traders. APMC Act needs to be replaced by the New Agriculture Produce and Livestock Marketing, (Promotion and Facilitation) (APLM) Act, 2017.
- *Principle 3:* Reducing post-harvest losses (PHL) in fruits and vegetables, which are approx. 40 per cent in India. Agri-logistics can help generate post-harvest specialised infrastructure needed for complete integration of cold-chain facilities. In order to increase the income gains to farmers, they should be involved in post-harvest logistics as partners through the Farmer Producer Organisation (FPO)/Village Producer Organisation (VPO)/cooperative society. This would directly benefit the farmers by providing them access to bigger markets (beyond the local market) and more remunerative prices.
- *Principle 4:* Linking the processing industry with farmers and FPOs. The processing of perishable agricultural products is low (around 2 per cent). In fact, farmers do not have access to the food processing markets and most of their produce they sell directly to APMC markets and private traders and thus get less than 1/14th of the price consumer pays to the chain of middlemen

operating between farmers and retail and food processors. The Operation Green focuses on direct access of food processing market by FPOs. To facilitate this linkage, Agricultural Produce and Livestock Contract Farming and Services (Promotion & Facilitation) Act, 2018 has been drafted and circulated to all states for replacing it with existing APMC Acts.

Agricultural markets despite some changes continue to face many challenges like licensing barriers, lack of integration of various markets, lack of marketing infrastructure, high incidence of marketing charges, high wastages in supply chain, etc. Market dynamics continued with the protectionist perspectives. The best protection to farmers is in promoting long-term market linkages, such that markets are connected across place, time and form with farms. The necessary condition for transferring remunerative prices to the farmers is a competitive environment that facilitates fair and transparent price discovery.

E-National Agriculture Market (e-NAM)

The government introduced a Central Sector Scheme for promotion of a NAM to bring about a transformation in agriculture marketing environment. A unified market can be best realised through a pan-India electronic platform which can facilitate the participation of buyers and sellers from all over the country. The e-NAM network was inaugurated on 14-April-2016. As of July 2019, all 585 markets have been integrated into the scheme and 42.18 lakh farmers and 89,199 traders have been registered on e-NAM portal with a turnover of Rs. 16, 163.1 crore from the trading of 63.17 lakh tons produce. The government intends to link 22,000 mandis across the country with e-NAM, by 2021-22. The vision of a full-fledged e-NAM is where all types of markets have inter-operability in communication, standards, systems, operating under a common regulatory framework.

13.6.4 Policy Initiatives to Ensure Remunerative Price to Farmers for their Produce: MSP Policy and Procurement

The success of green revolution journey predicated upon, *inter-alia*, market support in terms of Minimum Support Price (MSP) and Procurement Policies. MSP was intended to serve as a floor price and an assurance against risks that could arise from sharp falls in the market price. The Government notifies MSPs based on the recommendations of an independent body, called Commission for Agricultural Costs and Prices (CACP). CACP recommendation is based on A-2 concept of cultivation which comprises of input costs in cash + Imputed value of family labour.

Minimum support prices acquire value only when they are supplemented by a robust mechanism of procurement, whenever they are breached in the market on the negative side. It is opined that the farmers would gain better from a more robust system of procurement rather than increase in MSPs. The existing procurement mechanisms by the government are implemented under:

- Food Corporation of India (FCI) operations for Central Pool Procurement is made to meet buffer norms and for meeting targets of the public distribution system.
- Price Support Scheme (PSS) applicable in case of MSP notified crops. Intervention by GoI whenever market prices fall below MSP.

- Market Intervention Scheme (MIS): To support commodities, for which MSPs are not notified - fruits/vegetables/other horticulture products.
- Price Stabilisation Fund (PSF): A scheme to protect consumers from rising prices.

Enhanced MSP (= 1.5 times of Cost of Cultivation-A-2) in Union Budget, 2018 followed by announcement of PM-AASHA for Enforcement

Giving a major boost to the pro-farmer initiatives of the Government and in keeping with its commitment and dedication for the Annadata, the government launched a Scheme “Pradhan Mantri Annadata Aay Sanrakshan Abhiyan” (PM-AASHA) which aimed at ensuring remunerative prices to the farmers for their produce as announced in the Union Budget for 2018. The new Umbrella Scheme PM-AASHA comprises of Price Support Scheme (PSS) (for Pulses, Oilseeds and Copra), Price Deficiency Payment Scheme (PDPS) (Oilseeds and Pulses) and Pilot of Private Procurement and Stockiest Scheme (PPPS) (only for Oilseeds in 15 districts).

Warehousing

An efficient marketing system alone is not sufficient and cannot guarantee desired benefits to farmers unless it is supported by an efficient storage system for post-harvest crops. Warehousing allows farmers to balance their supply to markets and in the interim, enables them to avail finance to meet their immediate financial requirements. Warehousing availability, of suitable type and quality, thus makes it an important component of the agricultural marketing system. Three main agencies in the public sector engaged in building large scale storage/warehousing capacity viz., Food Corporation of India (FCI), Central warehousing Corporation (CWC) and State Warehousing Corporation (SWCs). The government also supports the private sector and Cooperatives in creating warehousing capacity under the ‘Agricultural Marketing Infrastructure sub scheme’ (erstwhile rural godown scheme).

13.6.5 Policy Initiatives to Improve Resource Efficiency/Reduction in Cost

The inputs (seed, fertilizers and pesticides, farm machinery, credit and labour) account for major component of cost in case of all selected crops. The effective management of these inputs and their efficient use would help in reducing the cost and in turn in enhancing the productivity and income of the farmers.

Soil Health Management – Soil Health Card (SHC) Scheme

Diagnostic surveys in the Indo-Gangetic Plains (IGP) indicate that in several high productivity areas of irrigated ecosystems, farmers use excessive fertilizer N to maintain the yield levels attained previously with relatively less fertilizer. Indiscriminate use of N fertilizers aggravates soil fertility depletion, and proves detrimental in terms of low nutrient use efficiency, poor quality of produce, enhanced susceptibility of crops to biotic and abiotic stresses, and a potential threat of groundwater pollution due to excessive leaching of nitrates beyond effective root zone.

Enhancing nutrient use efficiency will be the key for sustained agricultural production, lowering of unit cost of cultivation and for raising farm income in

the years to come. SHC Scheme, launched on 19 February, 2015 aims at issuing SHC to each one of the 140 million farm holdings at 2-3-year interval on a continuing basis. The SHC would include analysis of 12 soil parameters viz., potential for hydrogen (pH), Emulsifiable Concentrate (EC), Soil Organic (SOC), available primary nutrients (N, P, K), available secondary nutrient (S), and available micro-nutrients (Zn, Fe, Cu, Mn, Boron). Based on analyses for these values, fertilizer and soil amendment recommendations are to be formulated for three prominent crops each of kharif (monsoon) and rabi (dry) seasons.

Issue of subsidised fertilizers to farmers was linked to the recommendations made in soil health card. SHC involves assessment of soil fertility variation and suggesting nutrient prescriptions following the principle of 4R (right rate, right source, right time and right method). As on 31st March 2019, around 130 million SHCs issued.

To rationalise the use of fertilizers, government announced Direct Benefit Transfer (DBT) scheme under which subsidies will be released for those fertilizers issued against the SHCs. The SHC-based fertilizer recommendation is expected to enhance productivity and income. But timely availability of all fertilizers recommended in the SHC is a pre-requisite for adoption of SHCs.

Seed – A critical Input

Seed is a critical input for enhancing productivity of all agricultural and horticultural crops and plays a vital role in improving the income status of farmers. Use of quality seeds alone can increase productivity by 15-20 per cent, showcasing its importance in agriculture. Sustained increase in agricultural production and productivity necessarily requires continuous development of new and improved varieties as well as hybrids, suitable for 128 agro-climatic zones of the country. This also necessitates an efficient seed multiplication system, integrating plant breeding, trade and distribution, and finally the farmer using the seeds. One of the serious factors responsible for low use of quality seed is sale of spurious seed in the market.

In order to improve the quality of farm saved seeds (60-65 per cent), Seed Village Programme being implemented from year 2005-06 onwards needs to be upgraded for better monitoring. The targeted 500 number of Seed Processing and Godown at the Gram Panchayat level by 2022 will strengthen seed production system.

Pest Management

In India, the farmer's crop yield losses range from 15 to 25 per cent owing to the presence of weeds, pests, diseases and rodents. The Government enacted Insecticides Act (I.A.), 1968 regulates import, manufacture, sale, transport, distribution and use of pesticides so as to prevent risks to human beings, animals and matters connected therewith. Challenges such as Quality of pesticide, Optimum application of pesticide, Popularising Integrated Pest Management (IPM) techniques, Price of generic pesticides and monopolistic practices still need to be addressed. The problem of excessive pesticide usage can be addressed through multi-stakeholder participation. In addition to farmers, pesticide producers and sellers, middleman of vegetable trade and consumers should also be involved in diagnosis of the problems and designing of approaches.

Agricultural Mechanisation speeds up tasks, and helps bring judicious use of inputs like seeds, fertilizer and water, contributing to farm productivity, and is, therefore aid in enhancing farmers' income. Estimates indicate that agricultural mechanisation can contribute a cut in cultivation cost by 25 per cent and rise in productivity by 20 per cent, thereby affecting an increase in farm income, to the extent of 25-30 per cent.

The Government of India is promoting agricultural mechanisation on farms through a Sub Mission on Agricultural Mechanisation (SMAM) of Rs. 2000 crore, which started from April 2014 under the umbrella of National Mission on Agricultural Extension and Technology (NMAET). The objectives of this Mission are: increasing the reach of farm mechanisation to small and marginal farmers and regions with low farm power; promoting 'Custom Hiring Centres'; creating hubs for hi-tech and high value farm equipment; awareness through demonstration and capacity building and ensuring performance testing and certification of machines.

Check Your Progress 5

- 1) 'Crop diversification is the key for raising the farmers Income'. Do you agree with the statement? Critically examine the strategy envisaged by the Government for promoting crop diversification in India.

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- 2) In the context of the stated aim of doubling farmer's income, discuss the challenges involved in realising the same.

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- 3) What is MSP? Explain its importance to Indian agriculture and to consumers.

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- 4) Examine the problems associated with food grains procurement and their storage by the government in India.

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13.7 RELEVANCE OF NON-AGRICULTURAL ACTIVITIES IN DOUBLING OF FARMERS' INCOME

The agrarian structure of India has been experiencing a reduction in farm size and increase in marginalisation of holdings for the past several decades. The small land base of the Indian farmer is one of the major factors contributing to rural poverty. If agriculture were to be the sole source of income for small landholders, the majority of them would live much below the poverty line. A number of studies from developing countries have suggested that diversification of rural economy towards non-farm activities has considerable potential to augment farmers' income and reduce rural poverty. Diversification towards nonfarm activities overcomes the land constraint to income growth, enables the farmers to cope up with the shocks of crop failure and enhances their capacity to invest in productivity-enhancing agricultural inputs and technologies.

NABARD Survey shows that 35 per cent income of agricultural households is from cultivation, 34 per cent from wage labour, 16 per cent from salaries and 8 per cent from livestock and remaining 7 per cent from non-farm sectors. NABARD survey also provides that:

- Only 13 per cent of agricultural households have one single source of income.
- Around 50 per cent of these households have two sources,
- 29 per cent three sources and
- 9 per cent four sources.

It shows that agricultural households do not depend only on farm income but they depend on multiple sources for their livelihoods. Thus, both agriculture and non-agriculture are important for raising income of agricultural households. According to NSSO SAS Surveys and NABARD Survey, the income of households was classified according to four sources, viz. agriculture, agricultural wages, livestock and non-farm income. Further discussion on income of the farmers earned through sources other than cultivation is beyond the scope of this unit and therefore needs to be covered separately in the different unit.

Check Your Progress 6

- 1) Briefly discuss the role and relevance of non-agricultural activities in raising the farmers income in India.

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13.8 LET US SUM UP

In India, agriculture is a key sector in terms of its contribution to both employment and GDP.

The green revolution in cereal production, white revolution in milk production, green revolution in cotton production and the more recent diversification of production towards pulses, fruit and vegetables as well as meat and meat products, have been largely in response to evolving demand patterns driven by rising incomes and urbanisation. As a result, India has transformed from a food deficit country to a major exporter of agriculture and allied products. The strong growth in production has been sustained by an improved access to inputs such as fertilizers and seeds, increased irrigation coverage (including micro irrigation), greater reach of institutional credit, introduction of the Kisan Credit Card scheme – enabling a more timely access to credit – and the designation of agriculture for priority lending. The beginning of structural change is underway due to the involvement of the private sector, going beyond the mainly traditional or unorganised private players (including mandi traders, private mills, village brokers, traditional retailers) to organised private entities, such as agribusiness and large food-processing companies or supermarkets.

Despite these successes, challenges remain. Among them, the notable being, slow pace of structural transformation and low labour productivity, continuing fragmentation of operational holding, inefficient agro-food supply chains and retail sector, prevalence of informal channels of credit and the climate change. Although, Indian agriculture has increasingly become integrated with world markets, its trade share is still low compared to the share of India's total merchandise exports and imports as a per cent of India's GDP. Environmental pressures are also starting to loom large. To address these concerns, agricultural policies in India are designed and implemented by a complex system of institutions. States have Constitutional responsibility for many aspects of agriculture, but the Central government plays an important role by developing national approaches to policy and providing the necessary funds for implementation at the state level. Throughout the last decades, agricultural policies have sought to achieve food security, often interpreted in India as self-sufficiency, while ensuring remunerative prices to producers and safeguarding the interest of consumers by making supplies available at affordable prices.

13.9 TERM-END EXERCISES

- 1) The need for a Second Green Revolution is being experienced more than ever before. Do you agree? What are your views with respect to bringing revolution in the production of nutrition-rich crops like pulses, fruits and vegetables — which remained untouched in the first Green Revolution?
- 2) Discuss any two steps taken by the government in the direction of improving agricultural marketing system in India since independence.
- 3) “Agriculture sector appears to be adversely affected by the economic reform process.” Elucidate.
- 4) Discuss the impact of the Climate change on the Indian Agriculture. What are the initiatives taken by the Government to mitigate this impact?
- 5) How farming system play important role in doubling the farmers income? Also, explain the role of Non-farm income in doubling farmers’ income.
- 6) What are the Policy Initiatives taken by the government to ensure Remunerative Price to farmers for their Produce?

13.10 KEYWORDS

Crop Diversification	: Crop diversification refers to the addition of new crops or cropping systems to agricultural production on a particular farm taking into account the different returns from value-added crops.
HYV Seeds	: High Yielding Variety Seeds (HYV seeds) are seeds of better quality than normal quality seeds. They are resistant and have a high yielding potential against insects and diseases.
Institutional Credit	: Agriculture credits provided by government institutions are called institutional credit.
Marginalising of Landholdings	: A general feature of the size distribution of operational holdings is that the percentage of holdings decreases as the holding size increases.
Minimum Support Prices	: The minimum support price (MSP) is an agricultural product price, set by the Government of India to purchase directly from the farmer.

13.11 REFERENCES

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13.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Read Section 13.2 and answer
- 2) Read Sections 13.1 and 13.2 and answer
- 3) Read Sections 13.1 and 13.2 and answer

Check Your Progress 2

- 1) Read Section 13.3 and answer
- 2) Read Sub-section 13.3.1 and answer
- 3) Read Sub-section 13.3.5 and answer
- 4) Read Sub-section 13.3.4 and answer

Check Your Progress 3

- 1) Read Section 13.4 and answer
- 2) Read Sub-section 13.4.2 and answer
- 3) Read Section 13.4 and answer
- 4) Read Sub-section 13.4.3 and answer

Check Your Progress 4

- 1) Read Section 13.5 and answer

Check Your Progress 5

- 1) Read Sub-section 13.6.3 and answer
- 2) Read Section 13.6 and answer
- 3) Read Sub-section 13.6.4 and answer
- 4) Read Sub-section 13.6.5 and answer

Check Your Progress 6

- 1) Read Section 13.7 and answer

